

Joanna Cai

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EDUCATION

University of British Columbia

Sept 2025 - April 2029

Bachelor of Applied Science in Integrated Engineering - Mechanical and Electrical Concentration

Awards: 2025 TD National Scholar - \$70,000 CAD, Dr. Jim McEwen Innovation Award - \$5,000 CAD

TECHNICAL SKILLS

Software and Tools: Altium Designer, KiCad, SolidWorks, Fusion 360, Git, Oscilloscope, Multimeter, Figma

Languages: C/C++, Python, MATLAB, JavaScript, HTML/CSS

Hardware: ESP32, STM32, Raspberry Pi, CAN, UART, MIDI

EXPERIENCE

National AccessArts Centre

May 2026 - Present

Product Engineer Intern | ESP32, C++, Bluetooth MIDI, Altium PCB Design, SolidWorks, 3D Printing

Calgary, AB

- Developed a Bluetooth MIDI guitar strumming device that enables musicians with limited finger mobility to perform full chords by integrating an ESP32, IMU-based arm motion sensing, embedded C++, and a custom 3D-printed enclosure.
- Designed and assembled a custom USB MIDI controller PCB in Altium using a Raspberry Pi Pico, joystick, and tactile inputs to emulate multiple instruments and synthesizer controls through embedded firmware.
- Accelerated prototype development by designing, fabricating, and assembling custom enclosures and mechanical components using SolidWorks, 3D printing, soldering, and rapid prototyping workflows.
- Designed a 3D-printed paint tube roller to simplify paint extraction and uncapping for artists with limited hand mobility.

UBC Baja SAE Design Team

Sept 2025 – Present

Mechatronics Subteam Member | SolidWorks, Altium, C, CAN, STM32, Shop skills

Vancouver, BC

- Designed a custom LiDAR interface PCB integrating automotive 12 V regulation, transient voltage protection, and UART communication for RPLIDAR A1 sensor deployment.
- Implemented CAN communication on an STM32 microcontroller to distribute sensor data across the vehicle network.
- Designed and manufactured linear potentiometer mounts for suspension travel sensing, enabling real-time suspension telemetry over the vehicle CAN network.
- Designed and assembled the vehicle's primary wiring harness, integrating power, sensors, and custom PCBs into the vehicle electrical architecture.

PRIME Robotics Lab - Dr. Chao Liu

Sept 2025 - Present

UBC Undergraduate Research Assistant | Solidworks, C++

Vancouver, BC

- Designed a precision robotic gripper in SolidWorks for automated PCB handling by optimizing geometry for reliable grasping.
- Investigated tactile sensing, robotic manipulation, and machine learning techniques by evaluating literature and documenting design tradeoffs for future robotic systems.
- Collaborated with graduate researchers to analyze tactile, visual, and motion sensor data for human feedback replication in robotic manipulation.

Nokia

July 2024 - Aug 2024

Software Engineering Future Tech Intern | Python, Docker, Powerpoint, Jira, Confluence

Hybrid / Toronto, ON

- Increased API test coverage by developing automated testing frameworks using Pytest and Schemathesis for Nokia's Notifier API.
- Communicated testing results and technical recommendations to software and security teams, helping prioritize implementation and validation efforts.

PROJECTS

melokey | KiCad, Fusion 360, Figma, Swift

June 2024 - Present WIP

- Developed an accessible USB MIDI keyboard with a custom PCB, simplified key layout, adaptive software, and LED guidance to support neurodivergent learners.
- Designed a custom KiCad PCB and Fusion 360 enclosure, and prototyped a companion iOS app interface in Swift and Figma.

blipyboard | KiCad, Fusion 360, Circuit Python

May 2025 - June 2025

- Designed and built a custom macro keypad by developing a KiCad PCB, CircuitPython firmware, and a 3D-printed enclosure integrating a rotary encoder and OLED display.
- Implemented KMK firmware to support programmable macros, layer switching, and real-time OLED feedback, creating a fully customizable human-interface device.